

Every K-map we have drawn ended with a sum of products (SOP) expression. We grouped the 1s and ORed the resulting terms together. We can also use a K-Map to provide the product of sums (POS), where the final expression is AND-ed OR-terms.

- SOP $\rightarrow$ AND gates feeding into a single OR gate (AND-OR, two levels).
- POS $\rightarrow$ OR gates feeding into a single AND gate (OR-AND, two levels)

The 0s in a K-map represent the minterms of F' (the complement of F). If we group the 0s the same way we group 1s, we get a simplified SOP expression for F .

Grouping with 0s:

1. If a variable is 0 throughout the group $\rightarrow$ write it uncomplemented in the sum term.
2. If a variable is 1 throughout the group $\rightarrow$ write it complemented in the sum term.
3. If a variable varies across the group $\rightarrow$ it cancels; leave it out

Full K-Map Procedure:

1. Fill the map with 1s and 0s as usual.
2. For SOP: group the 1s $\rightarrow$ read each product term normally $\rightarrow$ OR them together.
3. For POS: group the 0s $\rightarrow$ read each sum term (with the flipped rule above) $\rightarrow$ AND them together.
4. Both results describe the same function.

Below is an example where you find the SOP and POS minimization for the same K-Map:

Simplify $F(A, B, C, D) = \sum(0, 1, 2, 5, 8, 9, 10)$ in both SOP and POS form.

Fill the map. 1s go in cells 0,1,2,5,8,9,10. All remaining cells get 0s.

		CD				
		00	01	11	10	
AB	00	1	1	0	1	<i>SOP</i> $\bar{B}\bar{C}$ $\bar{B}\bar{D}$ $\bar{A}\bar{C}D$
	01	0	1	0	0	
	11	0	0	0	0	
	10	1	1	0	1	

(a) SOP — group the 1s:

- **Blue solid** — m_0, m_1, m_8, m_9 (rows 00 and 10 wrap, cols 00 and 01): A varies, $B = 0 \rightarrow \bar{B}$, $C = 0 \rightarrow \bar{C}$, D varies. **Term:** $\bar{B}\bar{C}$
- **Cyan dashed** — m_0, m_2, m_8, m_{10} (four corners): A varies, $B = 0 \rightarrow \bar{B}$, C varies, $D = 0 \rightarrow \bar{D}$. **Term:** $\bar{B}\bar{D}$
- **Violet dotted** — m_1, m_5 (col 01, rows 00 and 01): $A = 0 \rightarrow \bar{A}$, B varies, $C = 0 \rightarrow \bar{C}$, $D = 1 \rightarrow D$. **Term:** $\bar{A}\bar{C}D$

$$F = \bar{B}\bar{C} + \bar{B}\bar{D} + \bar{A}\bar{C}D$$

(b) POS — group the 0s:

Now circle the 0s. The 0s are at $m_3, m_4, m_6, m_7, m_{11}, m_{12}, m_{13}, m_{14}, m_{15}$.

		CD				
		00	01	11	10	
AB	00	1	1	0	1	$B\bar{D}$
	01	0	1	0	0	CD
	11	0	0	0	0	AB
	10	1	1	0	1	

Read each 0-group using the **flipped rule**:

- **Red solid** — $m_{12}, m_{13}, m_{14}, m_{15}$ (row $AB = 11$): $A = 1 \rightarrow \bar{A}$, $B = 1 \rightarrow \bar{B}$, C varies, D varies. Sum term: $(\bar{A} + \bar{B})$
- **Orange dashed** — m_3, m_7, m_{11}, m_{15} (col $CD = 11$): A varies, B varies, $C = 1 \rightarrow \bar{C}$, $D = 1 \rightarrow \bar{D}$. Sum term: $(\bar{C} + \bar{D})$
- **Dotted** — m_4, m_6, m_{12}, m_{14} (wrap, $B = 1, D = 0$): A varies, $B = 1 \rightarrow \bar{B}$, C varies, $D = 0 \rightarrow D$. Sum term: $(\bar{B} + D)$

$$F = (\bar{A} + \bar{B})(\bar{C} + \bar{D})(\bar{B} + D)$$

Don't Care Conditions: an input combination that is unspecified, or whose output simply doesn't matter to us. For those inputs, we don't care whether the circuit outputs 0 or 1, so we are free to assign whichever value makes our simplification better.

Decimal Digit	BCD 8421
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
Unused bit combinations	1010
	1011
	1100
	1101
	1110
	1111

The truth tables we have done give outputs to every combination, but some circuits don't require every combination we can make. For example, BCD.

In 4-bit BCD circuits, the computer only cares about the input combinations for the digits 0-9. 10-15 are our don't care conditions (we cannot view them as single digits that we can create numbers with).

How to use Don't Cares:

1. Mark all minterms where $F = 1$ with a 1 as usual.
2. Mark all don't-care minterms with an X.
3. When forming groups, you may include any X in a group — treat it as a 1 if doing so makes the group larger (and therefore the term simpler).
4. You do not have to include every X. Only include the ones that help.
5. Every actual 1 must be covered. Xs that are not included in any group are simply treated as 0.

Don't-care conditions are listed separately from the function.

$$F(w, x, y, z) = \sum m(a, b, c, \dots) \quad d(w, x, y, z) = (p, q, r, \dots)$$

Simplify $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$ with don't-cares $d(w, x, y, z) = (0, 2, 5)$.

Fill the map. Place 1 in cells 1,3,7,11,15. Place X in cells 0,2,5. Everything else is 0.

		yz				
		00	01	11	10	
wx	00	X	1	1	X	$\bar{w}\bar{x}$
	01	0	X	1	0	yz
	11	0	0	1	0	
	10	0	0	1	0	

Without don't-cares: m_1 and m_3 would form a 2-cell group giving the 3-literal term $w'x'z$.

With don't-cares: by including Xs at m_0 and m_2 , we expand to a 4-cell group covering the entire top row, giving the 2-literal term $\bar{w}\bar{x}$. The X at m_5 is not needed for any group, so we leave it as 0.

Final answer:

$$F = yz + \bar{w}\bar{x}$$

Ex)

This problem is structured exactly like Question 5 on your homework.

Simplify $F(w, x, y, z) = \sum(3, 4, 5, 6, 7)$ with don't-care conditions $d(w, x, y, z) = (10, 11, 12, 13, 14, 15)$.

Fill the map. 1s at: 3, 4, 5, 6, 7. X at: 10, 11, 12, 13, 14, 15. 0s everywhere else (cells 0, 1, 2, 8, 9).

		yz				
		00	01	11	10	
wx	00	0	0	1	0	yz
	01	1	1	1	1	
	11	X	X	X	X	x
	10	0	0	X	X	

Strategy — think before you circle:

Before drawing any groups, notice that the entire row $wx = 11$ and two cells of $wx = 10$ are don't-cares. This is a huge advantage — we can use them to make our 1-groups much larger.

Group 1 (blue) — 8 cells, rows $wx = 01$ and $wx = 11$:

The entire $wx = 01$ row (real 1s: m_4, m_5, m_6, m_7) is expanded by including the entire $wx = 11$ row (Xs: $m_{12}, m_{13}, m_{14}, m_{15}$).

w varies, $x = 1$ always $\rightarrow x$, y varies, z varies. **Term:** x

Group 2 (red, dashed) — 4 cells, col $yz = 11$:

m_3 (real 1) combined with m_7 (real 1, reuse is fine), m_{11} (X), m_{15} (X).

w varies, x varies, $y = 1 \rightarrow y$, $z = 1 \rightarrow z$. **Term:** yz

Check coverage: every real 1 is covered. $m_3 \rightarrow$ Group 2. $m_4, m_5, m_6, m_7 \rightarrow$ Group 1. ✓

Xs used: $m_{12}, m_{13}, m_{14}, m_{15}$ (treated as 1). Xs not used: m_{10}, m_{11} (treated as 0). That's fine.

Final answer:

$$F = x + yz$$

Ex)